



cations that require data processing. It supports data encryption and decryption for security-conscious users and businesses handling data. PC OEMs can integrate the YRS820 into their systems for AI-capable devices.

Yingren

<http://www.yingren.com>

Efficient and powerful Edge AI accelerators

Hailo has introduced the Hailo-10 accelerator family which is designed to run generative AI models using less than



five watts of power. The company claims that these accelerators can provide up to 40 tera-operations per second (TOPS) for machine learning and AI tasks, offering double the performance of Intel's Core

Ultra NPU at half the power. The accelerators are designed for PC manufacturers to enhance machine learning and AI capabilities in personal computers, especially for generative AI applications. They are also intended for the automotive industry to provide advanced AI features in automotive infotainment systems. Developers and engineers working on edge AI applications that require low power consumption and high performance, such as chatbots and content generation tools, will also find these accelerators useful.

Hailo

<https://hailo.ai>

Micro-power Wi-Fi system for IoT connectivity

Qualcomm has launched QCC730, a Wi-Fi chip, designed for IoT devices with enhanced range, data transfer



rates, and 88% less power consumption per data transfer than its previous generation. The chip features direct cloud connectivity, Matter

integration, and operates in both hosted and hostless modes. The Wi-Fi chip is

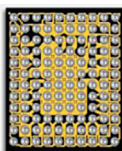
intended for use by manufacturers and developers of IoT devices that require enhanced range, improved data transfer rates, and reduced power consumption. This includes a wide range of applications, such as smart home devices, wearable technology, industrial sensors, and other connected devices that benefit from direct cloud connectivity and efficient power usage.

Qualcomm

<https://www.qualcomm.com>

'Neural decision' chip

Syntiant Corp. has launched the Syntiant NDP250 neural decision processor, delivering five times the tensor throughput of its predecessors. Utilising the Syntiant Core 3 programmable deep learning architecture, the NDP250 is



designed for applications in power-efficient devices,

such as person detection, object classification, automatic speech recognition (ASR), text to speech (TTS), and motion tracking. Targeted users include developers and manufacturers of AI-enabled devices. This encompasses consumer electronics companies producing smart home products like video doorbells and security cameras, automakers incorporating AI into vehicle security systems, IoT device developers, and manufacturers of mobile and wearable technology. Enterprises using systems like automated guided vehicles will also benefit from the processor's low-latency edge processing.

Syntiant Corp.

<https://www.syntiant.com>

Microcontrollers that enhance smart devices with AI

Alif Semiconductor has introduced the Balletto microcontroller family, featuring innovations such as Bluetooth Low Energy (BLE), support for the Matter standard, and a neural processing unit (NPU) accelerator aimed at enhancing

machine learning and artificial intelligence tasks. The microcontroller family targets developers and manufacturers across various industries. Consumer electronics companies will find them



useful for smart home devices and wireless audio systems due to BLE

and Matter support. Healthcare device manufacturers can leverage them for wearable health monitors and hearing aids, thanks to their efficient power use and processing capabilities. Additionally, these microcontrollers are suitable for industrial applications like point-of-sale systems and enterprise-level hardware, as well as in security technology and automotive industries, where reliable, secure, and high-performance embedded solutions are required.

Alif Semiconductor

<https://alifsemi.com>

Compact satellite module boosts connectivity

Quectel Wireless Solutions has launched BG95-S5 3GPP, a non-terrestrial network (NTN) satellite communication module. Designed with a compact SMT form factor of 23.6mm × 19.9mm × 2.2mm, the BG95-S5 is suitable for applications



with limited space that require access to multiple network technologies. As NTN satellite communication gains traction with expanding constella-

tions, this module offers the flexibility to connect via NTN, cellular, and GNSS, ensuring reliable connectivity regardless of location or the immediate availability of NTNs. The module also features low power consumption and an extended operating temperature range of -40°C to +85°C. It is also crucial for emergency and disaster management, smart city technology integrators, and machine-to-machine (M2M) communication applications.

Quectel Wireless Solutions

<https://www.quectel.com>